System Project Requirement No. 2

Database Schema

AniMAL - A Web-Based Market Access Platform for Farmers

Direct Farm-to-Buyer Transaction Platform

Submitted by:

CANTO, CHRISTIAN REYNALD

OCAMPO, RHIAN KATE CABATBAT

VILLACOTE, JUAN MIGUEL PORTOS

CORAL, KIAN CABURNAY

DIVINA, SAMUEL DEOCARES

Database Management Systems (DBMS) - Systems Project Requirement

1. System and Database Name

The central database instance is named animal_marketplace and runs on MySQL 8.0+ with the InnoDB storage engine. It also exposes two read-only views, v_active_listings and v_order_summary, that present frequently needed report data without duplicating storage.

2. Group Members

CANTO, CHRISTIAN REYNALD

OCAMPO, RHIAN KATE CABATBAT

VILLACOTE, JUAN MIGUEL PORTOS

CORAL, KIAN CABURNAY

DIVINA, SAMUEL DEOCARES

3. Table Summary

admin_audit_log - Before/after JSON snapshots of admin actions.

admin_order_holds - Admin intervention flags that pause an order until released.

buyer_profiles - Shopper records and profile keyed by user_id.

farmer_profiles - Verified seller records, location, valid ID, and verification workflow.

listing_images - Photos attached to a listing; primary flag for thumbnails.

order_status_history - Immutable log of order state transitions with actor and reason.

orders - Order header holding buyer, farmer, delivery, amounts, and current status.

produce_categories - Product taxonomy.

produce_listings - Sellable produce posts with price, stock, and lifecycle status.

quantity_units - Supported units of measure and conversion factors.

retention_holds - Legal/administrative holds preventing deletion of entities.

sessions - Active login sessions with device and IP fingerprint.

stock_adjustments - Audit trail of every stock change to a listing.

user_suspensions - Account suspension records and lift audit.

users - System accounts; single sign-on identity for all roles (farmer, buyer, admin).

Views: v_active_listings (active listings joined with farmer, category, and unit details); v_order_summary (order rollup with buyer, farmer, listing, and status).

4. Key Mapping

Primary keys: every table has a PK - surrogate AUTO_INCREMENT ids (users, produce_categories, quantity_units, produce_listings, listing_images, stock_adjustments, orders, order_status_history, admin_order_holds, admin_audit_log, user_suspensions, retention_holds), or a natural/single-column PK (sessions.id VARCHAR(128), farmer_profiles.user_id, buyer_profiles.user_id).

Foreign keys reference the table and column shown in the CONSTRAINT column of the schema below. Twenty-four FK references are declared: twenty-two columns marked FK plus the natural-key PK columns farmer_profiles.user_id and buyer_profiles.user_id, which are also foreign keys to users(id). ON DELETE semantics: CASCADE for child records tied directly to the parent identity (sessions, farmer_profiles, buyer_profiles, listing_images, stock_adjustments, order_status_history, admin_order_holds, user_suspensions, produce_listings.farmer_id); SET NULL for nullable audit attribution (verified_by, released_by, lifted_by); RESTRICT (default) for protected business data (category_id, unit_id, created_by, changed_by, admin_id, and orders.buyer_id/farmer_id/listing_id).

5. Normalization Rationale (3NF)

1NF: All column values are atomic and there are no repeating groups. Multi-valued data (listing photos, order status history, stock changes) is moved into separate child tables, and every table has a primary key.

2NF: No partial dependencies on the primary key. Profile tables that are naturally keyed by user_id (farmer_profiles, buyer_profiles) depend on the whole of that key, and every other table uses a single-column surrogate PK.

3NF: No transitive dependencies. Unit metadata (symbol, type, base_factor) lives in quantity_units and is referenced by produce_listings.unit_id; frozen order pricing and fees are stored directly on orders (unit_price, delivery_fee) so history does not depend on the volatile produce_listings.price_per_unit.

6. Database Schema (Table / Column / Data Type / Key / Null / Constraints)

Table	Column	Data Type	Key	Null	Constraints
users	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
users	email	VARCHAR(255)	UQ	No	UNIQUE
users	password_hash	VARCHAR(255)	-	No	-
users	role	ENUM('farmer','buyer','admin')	-	No	DEFAULT 'buyer'
users	status	ENUM('active','suspended','pending_verification')	-	No	DEFAULT 'pending_verification'
users	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP

Table	Column	Data Type	Key	Null	Constraints
users	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
users	last_login_at	TIMESTAMP	-	Yes	-
sessions	id	VARCHAR(128)	PK	No	-
sessions	user_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE CASCADE
sessions	ip_address	VARCHAR(45)	-	No	-
sessions	user_agent	TEXT	-	No	-
sessions	payload	TEXT	-	No	-
sessions	last_activity	INT UNSIGNED	-	No	-
farmer_profiles	user_id	BIGINT UNSIGNED	PK	No	References users(id) ON DELETE CASCADE
farmer_profiles	farm_name	VARCHAR(150)	-	No	-
farmer_profiles	farm_description	TEXT	-	Yes	-
farmer_profiles	contact_person	VARCHAR(100)	-	No	-
farmer_profiles	contact_phone	VARCHAR(20)	-	Yes	-
farmer_profiles	contact_email	VARCHAR(255)	-	Yes	-
farmer_profiles	address	TEXT	-	No	-
farmer_profiles	barangay	VARCHAR(100)	-	No	-
farmer_profiles	municipality_city	VARCHAR(100)	-	No	-
farmer_profiles	province	VARCHAR(100)	-	No	-
farmer_profiles	region	VARCHAR(100)	-	No	-
farmer_profiles	latitude	DECIMAL(10,8)	-	Yes	-
farmer_profiles	longitude	DECIMAL(11,8)	-	Yes	-
farmer_profiles	valid_id_type	ENUM('national_id','drivers_license','passport','voters_id','postal_id','senior_citizen_id','pwd_id')	-	No	-
farmer_profiles	valid_id_number	VARCHAR(50)	-	No	-
farmer_profiles	valid_id_image	VARCHAR(500)	-	Yes	-
farmer_profiles	verification_status	ENUM('pending','approved','rejected')	-	No	DEFAULT 'pending'
farmer_profiles	verified_at	TIMESTAMP	-	Yes	-
farmer_profiles	verified_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL
farmer_profiles	rejection_reason	TEXT	-	Yes	-
farmer_profiles	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
farmer_profiles	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
buyer_profiles	user_id	BIGINT UNSIGNED	PK	No	References users(id) ON DELETE CASCADE
buyer_profiles	full_name	VARCHAR(150)	-	No	-
buyer_profiles	contact_phone	VARCHAR(20)	-	Yes	-

Table	Column	Data Type	Key	Null	Constraints
buyer_profiles	delivery_address	TEXT	-	Yes	-
buyer_profiles	barangay	VARCHAR(100)	-	Yes	-
buyer_profiles	municipality_city	VARCHAR(100)	-	Yes	-
buyer_profiles	province	VARCHAR(100)	-	Yes	-
buyer_profiles	region	VARCHAR(100)	-	Yes	-
buyer_profiles	latitude	DECIMAL(10,8)	-	Yes	-
buyer_profiles	longitude	DECIMAL(11,8)	-	Yes	-
buyer_profiles	preferred_pickup	TINYINT(1)	-	No	DEFAULT 1
buyer_profiles	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
buyer_profiles	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
produce_categories	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
produce_categories	name	VARCHAR(100)	UQ	No	UNIQUE
produce_categories	slug	VARCHAR(100)	UQ	No	UNIQUE
produce_categories	description	TEXT	-	Yes	-
produce_categories	sort_order	INT	-	No	DEFAULT 0
produce_categories	is_active	TINYINT(1)	-	No	DEFAULT 1
produce_categories	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
produce_categories	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
quantity_units	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
quantity_units	name	VARCHAR(50)	UQ	No	UNIQUE
quantity_units	symbol	VARCHAR(10)	UQ	No	UNIQUE
quantity_units	type	ENUM('weight','count','volume')	-	No	DEFAULT 'weight'
quantity_units	base_factor	DECIMAL(10,4)	-	No	DEFAULT 1.0000
quantity_units	step	DECIMAL(10,4)	-	No	DEFAULT 0.01
quantity_units	is_active	TINYINT(1)	-	No	DEFAULT 1
produce_listings	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
produce_listings	farmer_id	BIGINT UNSIGNED	FK	No	References farmer_profiles(user_id) ON DELETE CASCADE
produce_listings	category_id	BIGINT UNSIGNED	FK	No	References produce_categories(id) ON DELETE RESTRICT
produce_listings	unit_id	BIGINT UNSIGNED	FK	No	References quantity_units(id) ON DELETE RESTRICT
produce_listings	name	VARCHAR(150)	-	No	-
produce_listings	description	TEXT	-	Yes	-
produce_listings	price_per_unit	DECIMAL(12,2)	-	No	-
produce_listings	available_quantity	DECIMAL(12,4)	-	No	DEFAULT 0.0000
produce_listings	minimum_order_q	DECIMAL(12,4)	-	No	DEFAULT 0.0100

Table	Column	Data Type	Key	Null	Constraints
	ty				
produce_listings	maximum_order_qty	DECIMAL(12,4)	-	Yes	-
produce_listings	harvest_date	DATE	-	Yes	-
produce_listings	expiry_date	DATE	-	Yes	-
produce_listings	is_organic	TINYINT(1)	-	No	DEFAULT 0
produce_listings	status	ENUM('draft','active','paused','sold_out','archived')	-	No	DEFAULT 'draft'
produce_listings	rejection_reason	TEXT	-	Yes	-
produce_listings	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
produce_listings	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
produce_listings	published_at	TIMESTAMP	-	Yes	-
listing_images	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
listing_images	listing_id	BIGINT UNSIGNED	FK	No	References produce_listings(id) ON DELETE CASCADE
listing_images	image_path	VARCHAR(500)	-	No	-
listing_images	is_primary	TINYINT(1)	-	No	DEFAULT 0
listing_images	sort_order	INT	-	No	DEFAULT 0
listing_images	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
stock_adjustments	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
stock_adjustments	listing_id	BIGINT UNSIGNED	FK	No	References produce_listings(id) ON DELETE CASCADE
stock_adjustments	adjustment_type	ENUM('initial','restock','sale_deduction','return','spoilage','admin_correction')	-	No	-
stock_adjustments	quantity_change	DECIMAL(12,4)	-	No	-
stock_adjustments	quantity_before	DECIMAL(12,4)	-	No	-
stock_adjustments	quantity_after	DECIMAL(12,4)	-	No	-
stock_adjustments	reference_type	ENUM('order','manual','admin')	-	Yes	-
stock_adjustments	reference_id	BIGINT UNSIGNED	-	Yes	-
stock_adjustments	notes	TEXT	-	Yes	-
stock_adjustments	created_by	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
stock_adjustments	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
orders	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
orders	order_number	VARCHAR(30)	UQ	No	UNIQUE
orders	buyer_id	BIGINT UNSIGNED	FK	No	References buyer_profiles(user_id) ON DELETE RESTRICT
orders	farmer_id	BIGINT UNSIGNED	FK	No	References farmer_profiles(user_id) ON DELETE RESTRICT

Table	Column	Data Type	Key	Null	Constraints
orders	listing_id	BIGINT UNSIGNED	FK	No	References produce_listings(id) ON DELETE RESTRICT
orders	quantity	DECIMAL(12,4)	-	No	-
orders	unit_price	DECIMAL(12,2)	-	No	-
orders	delivery_fee	DECIMAL(12,2)	-	No	DEFAULT 0.00
orders	delivery_method	ENUM('pickup','farmer_delivery')	-	No	DEFAULT 'pickup'
orders	delivery_address	TEXT	-	Yes	-
orders	preferred_pickup_date	DATE	-	Yes	-
orders	preferred_pickup_time	TIME	-	Yes	-
orders	buyer_note	TEXT	-	Yes	-
orders	farmer_note	TEXT	-	Yes	-
orders	admin_note	TEXT	-	Yes	-
orders	status	ENUM('pending','delivery_fee_review','confirmed','preparing','ready_for_pickup','out_for_delivery','completed','rejected','cancelled')	-	No	DEFAULT 'pending'
orders	rejection_reason	TEXT	-	Yes	-
orders	cancellation_reason	TEXT	-	Yes	-
orders	cancelled_by_role	ENUM('buyer','farmer','admin')	-	Yes	-
orders	confirmed_at	TIMESTAMP	-	Yes	-
orders	preparing_at	TIMESTAMP	-	Yes	-
orders	ready_at	TIMESTAMP	-	Yes	-
orders	out_for_delivery_at	TIMESTAMP	-	Yes	-
orders	completed_at	TIMESTAMP	-	Yes	-
orders	rejected_at	TIMESTAMP	-	Yes	-
orders	cancelled_at	TIMESTAMP	-	Yes	-
orders	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
orders	updated_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
order_status_history	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
order_status_history	order_id	BIGINT UNSIGNED	FK	No	References orders(id) ON DELETE CASCADE
order_status_history	from_status	ENUM('pending','delivery_fee_review','confirmed','preparing','ready_for_pickup','out_for_delivery','completed','rejected','cancelled')	-	Yes	-

Table	Column	Data Type	Key	Null	Constraints
order_status_history	to_status	ENUM('pending','delivery_fee_review','confirmed','preparing','ready_for_pickup','out_for_delivery','completed','rejected','cancelled')	-	No	-
order_status_history	changed_by	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
order_status_history	changed_by_role	ENUM('buyer','farmer','admin')	-	No	-
order_status_history	reason	TEXT	-	Yes	-
order_status_history	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
admin_order_holds	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
admin_order_holds	order_id	BIGINT UNSIGNED	FK	No	References orders(id) ON DELETE CASCADE
admin_order_holds	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
admin_order_holds	reason	TEXT	-	No	-
admin_order_holds	is_active	TINYINT(1)	-	No	DEFAULT 1
admin_order_holds	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
admin_order_holds	released_at	TIMESTAMP	-	Yes	-
admin_order_holds	released_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL
user_suspensions	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
user_suspensions	user_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE CASCADE
user_suspensions	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
user_suspensions	reason	TEXT	-	No	-
user_suspensions	suspended_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
user_suspensions	lifted_at	TIMESTAMP	-	Yes	-
user_suspensions	lifted_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL
retention_holds	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT
retention_holds	entity_type	ENUM('user','order','listing','verification')	-	No	-
retention_holds	entity_id	BIGINT UNSIGNED	-	No	-
retention_holds	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
retention_holds	reason	TEXT	-	No	-
retention_holds	expires_at	TIMESTAMP	-	Yes	-
retention_holds	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP
retention_holds	released_at	TIMESTAMP	-	Yes	-
retention_holds	released_by	BIGINT UNSIGNED	FK	Yes	References users(id) ON DELETE SET NULL
admin_audit_log	id	BIGINT UNSIGNED	PK	No	AUTO_INCREMENT

Table	Column	Data Type	Key	Null	Constraints
admin_audit_log	admin_id	BIGINT UNSIGNED	FK	No	References users(id) ON DELETE RESTRICT
admin_audit_log	action	VARCHAR(100)	-	No	-
admin_audit_log	entity_type	VARCHAR(50)	-	No	-
admin_audit_log	entity_id	BIGINT UNSIGNED	-	Yes	-
admin_audit_log	old_values	JSON	-	Yes	-
admin_audit_log	new_values	JSON	-	Yes	-
admin_audit_log	ip_address	VARCHAR(45)	-	Yes	-
admin_audit_log	user_agent	TEXT	-	Yes	-
admin_audit_log	created_at	TIMESTAMP	-	No	DEFAULT CURRENT_TIMESTAMP